

End-to-End Snowflake Project

“RetailHub Sales Analytics Modernization”

This project covers almost all major Snowflake concepts in only 10 practical steps.

Project Overview

RetailHub wants to migrate on-premises retail sales data into Snowflake for:

- Reporting
- Analytics
- Real-time ingestion
- Cost optimization
- Secure data sharing

Data Sources:

- CSV sales files
 - JSON mobile logs
 - AWS S3 files
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Project Architecture

CSV / JSON Files

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Internal Stage / AWS S3

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Snowpipe / COPY INTO

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Bronze Layer

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Streams & Tasks

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Silver Layer

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Dynamic Tables

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Gold Layer

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Views / Data Sharing

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Dashboards

Step 1: Environment Setup

Create:

- Database
- Schemas
- Warehouse

Warehouse Configuration:

- SMALL size
- AUTO_SUSPEND = 60
- AUTO_RESUME = TRUE

Step 2: Create Internal Stage & File Format

Create:

- Internal stage
- CSV file format

Upload files using SnowSQL.

Step 3: Create Tables & Load Data

Create raw sales table:

- order_id
- customer_name
- product
- amount
- order_date

Load data using:

```
COPY INTO SALES_RAW  
FROM @RETAIL_STAGE;
```

Step 4: Implement Medallion Architecture

Create:

- Bronze Layer → Raw data
 - Silver Layer → Cleaned data
 - Gold Layer → Business aggregates
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Step 5: Perform Transformations

Transform sales data:

- Update product prices
 - Add STATUS column
 - Clean NULL values
 - Aggregate sales
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Step 6: Configure Streams, Tasks & Dynamic Tables

Create:

- Streams for CDC
 - Tasks for scheduled processing
 - Dynamic Tables for auto-refresh
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Step 7: AWS S3 Integration & Snowpipe

Create:

- External stage using AWS S3
- Snowpipe for automatic loading

Workflow:

AWS S3 → Snowpipe → Bronze Tables

Step 8: Semi-Structured Data & Optimization

Load JSON logs into VARIANT columns.

Apply:

- Clustering
- Auto Clustering

Check:

```
SELECT SYSTEM$CLUSTERING_INFORMATION('SALES_RAW');
```

Step 9: Time Travel, Cloning & Resource Monitors

Recover deleted data using Time Travel.

Create QA clone:

```
CREATE DATABASE RETAIL_QA CLONE RETAIL_DW;
```

Monitor warehouse credits using Resource Monitor.

Step 10: Security, Views & Data Sharing

Create:

- Roles
- Secure Views
- Materialized Views
- Data Shares

Provide dashboard access to business users.
